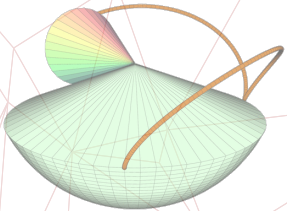

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The Value of Business Intelligence

Business Intelligence and Data Visualization

Program Core

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Introduction

- According to the PricewaterhouseCoopers Global Data Management Survey of 2001, *"Companies that manage their data as a strategic resource and invest in its quality are already pulling ahead in terms of reputation and profitability."*
- According to a traditional view, data is the "fuel" driving the automation of a business operation, implying that a company uses computers to help run its business.
- The forward-looking view, the strategic knowledge is embedded in the collection of a company's data and that extracting actionable knowledge will help a company improve its business.
- Consider this: Is a supermarket chain a business that sells food, *or*
- Is it a business that exploits knowledge about customer preferences, geographical biases, supply chain logistics, product life cycle, and competitive sales information to optimize its delivery, inventory, pricing, and product placement as a way to increase margin for each item sold.

Introduction

- The answer to that question may ultimately determine your company's long-term viability in the Information Age.
- So, how do you transform data into a strategic resource?
- It involves properly applying new technology to your data, but the most important part is being able to understand and subsequently build the business case for the value of information.
- In this chapter we look at the difference between the traditional use of data in a transactional environment and the modern view of data as a valuable resource that can be used for analytical purposes.
- We discuss what the aspects of information are that make up this value, and what kinds of processing can be performed to add value to data.

Information Asset and Data Valuation

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Information Asset and Data Valuation

- There are some reasons to consider data both ways i.e., asset and liability.
- If all a company does is accumulate and store data, there is some cost associated with the ongoing management of that data–the cost of storage, maintenance, office space, support staff, etc.
- This should show up on the balance sheet as a liability.
- Alternatively, data can be viewed as an asset, because data can be used to provide benefits to the company, is controlled by the organization, and is the result of previous transactions.
- But, organizations do not treat data as an asset; for example, there is no depreciation schedule for purchased data.
- Treating data as an asset is important, though, because it allows us to build the business case for investing in BI.
- We must have some way to measure the value of data. There are relatively few situations where we can accurately assign a discrete price for information.

Information Asset and Data Valuation

- This pricing structure is more frequently value based than if data were treated as a commodity.
- As an example, the telephone company charges you, say, $1 for each directory assistance inquiry, although that same telephone number could be acquired free through an online directory.
- The difference in cost is based on the convenience value of being able to pick up the receiver and get the number right away.
- The value of information depends on a number of contributing factors.
- As we are made aware of these factors, we can get a lot closer to developing a model for information valuation.
- Treating data as an asset is important, though, because it allows us to build the business case for investing in BI.
- This is not to say that we can accurately enumerate data as an asset on the balance sheet, but it does give some parameters to understanding the value of information and, consequently, the value of BI.

I. The Time Value of Data

- There is a timeliness or currency component to the value of information.
- Here's a simple example: If it is March 1st and some prophet told you today that for a certainty the March 2nd closing price of IBM stock is going to be $10 higher than the March 1st price.
- You can exploit that piece of information today to buy as much IBM stock as you can and sell it at the $10 profit on March 2nd.
- Yet if I gave you the same information on March 3rd, you could not exploit it in the same way.
- In this case, a large portion of the value of that piece of information is bound to its timeliness.
- It is clear that a portion of the value of information is related to time and that value may degrade as time elapses.

I. The Time Value of Data

- Stored data represents a snapshot of a real-world state at a particular point in time, then in the absence of continuous maintenance, as the world changes our snapshot grows more out of synch with reality.
- For example, our direct mail database may be of high value at its initial creation, but because it is estimated that about 20% of the population changes addresses each year, then if that database is not updated with new addresses, its value declines with time.
- Bad data actually can be viewed as a liability: There is no reduction in the fixed costs associated with managing that bad data.
- Using the data as part of an integration or linkage process will yield incorrect answers, and the value of the data set as a whole declines.

II. Information as a Shareable Resource

- Information is shareable, and the value of information increases as more people use it.
- An example of this is the knowledge of a process for sales professionals to alert them to the best time to contact a prospect.
- If that salesperson were to share that knowledge with other members of the sales staff, there is no degradation in the value that can be achieved by any one of the individuals aware of that knowledge.
- This means that the value of that information is multiplied by the number of people who know it.
- In the BI space, this is manifested through the data warehouse, which is used as a central repository for large amounts of shared data.
- If there is some economic value to be derived from a piece of information, that value can be increased through sharing.

III. Increasing Value through Increased Use

- For most assets, as usage increases, there is some depreciation in asset value. For example, a building's cost decreases as it gets older.
- On the other hand, the value of data does not decrease with use, because there is no degradation in information based on the number of times it is viewed.
- When everyone in the organization knows what information is available, how to access that information, and how to exploit that information, the value of that information rapidly increases.
- If data is stored, managed, and never used, there is no added value. So, it becomes a liability.

IV. Increasing Value through Quality

- The forecaster is a false one and tells you that the stock price will fall $10 by the close of business tomorrow, although it really will rise $10.
- The way you will have exploited what you believe to be true will ultimately result in a significant loss of value instead of an increase.
- This highlights the value of accurate information. Also defining quality metrics and measuring them using those metrics.
- Having some understanding of the measure or quality of data being used for a decision support process.
- Let the decision maker determine the risks associated with relying on that data.

V. Increasing Value through Merging

- The process of combining bits of knowledge provides significant leverage when increasing the value of information.
- Having sales channel information is of value; having supply channel information is valuable.
- Combining supply channel information with sales channel information provides knowledge about the movement of products from supplier to customer.
- Information increases in value when it can be used to enhance and expand other pieces of information.
- The BI process revolves around the ability to collect, aggregate, and, most importantly, leverage the integration of different data sets together.
- There is a large increase in the value of information if it can be used as leverage in increasing an actionable knowledge base.
- In other words, if we can take two pieces of information, link them together, and infer something new that could not have been learned independently.

VI. Value Versus Volume

- Businesses do not necessarily gain greater value by having more information.
- The sheer amount of information that is produced every year is almost unbelievable; trying to integrate that with what has existed before seems like a gargantuan effort.
- The complexity of data integration grows steeply as the number of data sources is increased.
- There is a perception that the more information there is, the better. So, there is some fine line between having the right amount of information and having too much.
- Having the right amount implies that this information supports the defined business requirements.
- So that, you are able to integrate that information and provide and present it within the required time.
- There is a qualitative difference between having lots of data that comes from disparate data sources and having lots of data that derives from the same source.
- For example, maintaining a large amount of historical transaction data (such as point-of-sale data or call detail records) may prove to be more valuable when it comes to analyzing trends over longer periods of time.

VII. Measuring the Value of Information

- One way to assess the value of information is to look at some traditional valuation models for other assets.
- **Historical Cost:** In this method, we assess the value based on what had been paid to acquire or create the information or based on how much it would cost to replace the information.
- **Market value:** In this method, we assess the value based on what someone else would be willing to pay for the information.
- Data aggregators and packagers create products based on this model.
- When it comes to improving the quality of, improving the accessibility of, or enhancing information that is typically hard for individuals to acquire on their own.
- **Utility Value:** In this method, we assess the value of information based on the expected value to be derived from the information.

Actionable Knowledge - Return on Investment

- An asset retains its value only if we do something with it.
- Some investment is probably required to build the environment where data can be turned into knowledge, but the real benefit occurs when that knowledge is actionable.
- That means that an organization cannot just provide for the information factory; it must also have some methods for extracting value from that knowledge.
- To have identified actionable knowledge is one thing, but to take the proper action requires a nimble organization with individuals empowered to take that action.
- The components of this strategy include analyzing costs, increases in revenues that are related to the activity, and other distinguishable benefits.

1. **Fixed Costs:** Initial expenses for BI tools (e.g., database, reporting tools).
2. **Variable Costs:** Additional costs for special software components.
3. **Ongoing Costs:** Maintenance and operational expenses.

Actionable Knowledge - Return on Investment

4. **Benefit Value:** Benefits gained from using the BI insights.
5. **Costs and Benefits of Related Components:** Consider the expenses and advantages of other BI tools and systems that support this activity.
6. **Value Model:** Expected value outcome from the activity.
7. **Success Probabilities:** Likelihood of achieving the expected value from the actions.
8. **Break-Even and Profitability:** Time to recover costs and profit potential.

Example

Building a CRM data warehouse for the purpose of increasing the lifetime value of each customer within a company's customer base.

- The goal is to build a data warehouse that encapsulates all the data related to each individual customer.
- Building the data warehouse involves costs for hardware, a database system, software tools, and integration into the business.

Actionable Knowledge - Return on Investment

- Next there are the additional costs associated with the design and implementation of the warehouse model(s), as well as identifying the data sources.
- The expected benefit of the data warehouse is a 30% increase in each customer's lifetime value by the end of the third year following the launch of the data warehouse into production.
- The ROI model must offset the costs just described with the overall benefit value associated with the increase in lifetime value.
- If there is no break even point, the cost to build the data warehouse is more than the value derived from it.
- In that case, it is probably worth looking for additional value that can be derived from the project before pitching it to senior management.

Business Intelligence Applications

- The different uses of data and the contexts of each use as it pertains to the exploitation of information.
- For the most part, we can break those into two areas. The first area is operational data use, and the other is strategic use.
- The predominant use of information today is operational: how data helps run the business, as opposed to strategic information use, which helps improve the business.
- Clearly these both are valuable, and without the operational use of information a business could not survive.
- To determine the extent of the value to be derived from the strategic use of information as well as what strategic uses are of importance.
- In this section we review some of the strategic uses of information as manifested through BI analytics.

Business Intelligence Applications

I. Customer Analysis

- CRM, or Customer Relationship Management, uses various customer analytics to help businesses understand their customers better and maximize the value of each relationship.
- Following are different aspects of customer analytics that benefit the sales, marketing, and service organizations as they interact with the customers.

I.a. Customer Profiling

- The bulk of marketing traditionally casts a wide net and hopes to capture as many individuals as possible.
- Companies are realizing that all customers are not clones of some predefined market segment but are thinking individuals.
- Customer analytics involve constantly updating individual customer profiles with demographic, psychographic, and behavioral data. This helps businesses better understand each customer's preferences and needs.

I.b. Targeted Marketing

- Knowledge of a set of customer likes and dislikes can augment a marketing campaign to target small clusters of customers that share profiles.
- In fact, laser-style marketing is focused directly at individuals as a byproduct of customer analytics.

I.c. Personalization

- As more business moves online, the browser acts as a proxy for the company's first interface with the customer.
- Personalization, which is the process of crafting a presentation to the customer based on that customer's profile.
- Web site personalization exploits customer profiles to dynamically collect content designed for an individual, and it is meant to enhance that customer's experience.

I.d. Collaborative Filtering

- We have all seen e-commerce Web sites that suggest alternate or additional purchases based on other people's preferences.
- In other words, the information on a Web page may suggest that *"people who have purchased product X also have purchased product Y."*
- These kinds of suggestions are the result of a process called collaborative filtering, which evaluates the similarity between the preferences of groups of customers.
- This kind of recommendation generation creates relatively reliable cross-sell and up-sell opportunities.

I.e. Customer Satisfaction

- Another benefit of the customer profile is the ability to provide customer information to the customer satisfaction representatives.
- This can improve these representatives' ability to deal with the customer and expedite problem resolution.

I.e. Customer Lifetime Value

- A company determines who their best customers are by analyzing data such as purchase frequency, spending amounts, and customer loyalty.
- The lifetime value of a customer is a measure of a customer's profitability over the lifetime of the relationship.
- It incorporates the costs associated with managing that relationship and the revenues expected from that customer.
- Customer analytics incorporates metrics for measuring customer lifetime value.

I.f. Customer Loyalty

- It is said that a company's best new customers are its current customers.
- This means that a company's best opportunities for new sales are with those customers that are already happy with that company's products or services.

II. Human Capital Productivity Analysis

- One way to attain value internally from BI is to be able to streamline and optimize people within the organization.

II.a. Call Center Utilization and Optimization

- If you have ever dawdled while on hold, waiting for a customer service representative to pick up the telephone.
- You can understand the value of analyzing call center utilization to look for ways to improve throughput and decrease customer waiting time.
- When a company's management realizes that inbound calls are likely to be from unsatisfied customers, making them stew on the phone is not going to improve customer satisfaction.
- In the more advanced cases, quick access to customer profile information may also affect the level of support provided to each customer (e.g., high level to high-value customers, minimal support to low-value customers).

II.b. Production Effectiveness

- This includes evaluating on-time performance, labor costs, production yield, etc. all as factors of how staff members' work.

III. Business Productivity Analysis

- Another popular analytic realm involves business productivity metrics and analysis.
- This includes various aspects such as performance measurement, efficiency evaluation, and productivity enhancement.

III.a. Defect Analysis

- While companies struggle to improve quality production, there may be specific factors that affect the number of defective items produced.
- This includes factors like the time of day, the source of raw materials, and the individuals staffing the production line. These factors can be exposed through one component of business productivity analytics.

III.b. Capacity Planning and Optimization

- Understanding resource utilization for all aspects of a physical plant i.e., all aspects of the machinery, personnel, expected throughput, raw input requirements, warehousing, just-in-time production, etc.
- A BI analytics process can assist management in resource planning and staffing.

III.c. Financial Reporting

- Stricter industry regulatory constraints may force companies to provide documentation about their financial.
- Especially now, as companies are failing because of misstated or inaccurate results.
- In addition, financial reporting analytics provide the means for high-level executives to take the pulse of the company and drill down on particular areas.

III.d. Capacity Planning and Optimization

- Having greater accuracy or precision in tracking business processes and productivity allows a manager to make better decisions about how and when to allocate resources in a way that minimizes risk to the organization.
- In addition, risk analysis can be factored into business decisions regarding the kind of arrangements that are negotiated with partners and suppliers.

III.e. Just-in-Time

- The concept of just-in-time product development revolves around the mitigation of inventory risk associated with commodity products with high price volatility.
- For example, the commodity desktop computer business is driven by successive generations of commodity components (disk drives, CPUs, DRAM memory chips, to name a few).

- To alleviate this, the knowledge of how quickly the production team can assemble a product, along with sales channel information and supplier information.
- It can help in accurately delivering products built to customer order within a predictable amount of time.

III.f. Asset Management and Resource Planning

- Utilization, productivity, and asset life-cycle information can be integrated through business analytics to provide insight into short- and long-term resource planning.
- As well as exposing optimal ways to manage corporate assets to support the resource plan.

IV. Sales Channel Analytics

- Sales channel analytics might be considered a subset of business productivity analytics.

Business Intelligence Applications

IV.a. Marketing

- Both the ability to fine-tune a marketing program and the ability to determine marketing effectiveness can be derived through sales channel analytics.
- A typical iterative process would be to identify a marketing strategy based on an analysis of a clustering of customers by profile and then to implement that strategy.
- The effectiveness of the strategy will ripple through the sales channel data, which can then be used to compare the actual results with expectations.
- The degree to which those expectations are met (or exceeded) can be feedback into the analytical processing to help determine new strategies.

IV.b. Sales Performance and Pipeline

- Data associated with the sales staff can be analyzed to identify variables that affect the efficiency of the sales cycle.
- Such as individual sales staff member, region, industry, contact people, contact times, and contact frequency.

V. Supply Channel Analytics

- Supply channel analytics are used to characterize and benchmark a company's supply channels from its various vendors and suppliers through internal inventory management.

V.a. Supplier and Vendor Management

- Many organizations are unable to identify who their vendors are or how many vendors are supplying products or services.
- Supply chain analytics allow a company's management to track performance and reliability by supplier.
- Evaluating and rating the quality of the products supplied, as well as help to optimize supplier relationships with respect to spending, procurement, and risk.

V.b. Shipping

- There are different methods by which a company delivers its products to its customers, each with its own cost schedule.
- For example, it may be more expensive to ship products by air than by truck, but the products will arrive at the destination faster if shipped by air.
- A company can minimize its delivery costs by being able to select the most efficient delivery method for any specific business.
- Also knowing whether the products can be available within the right time schedule, especially if your production depends on external suppliers.

V.c. Inventory Control

- Maintaining an inventory of commodity products that exhibit volatile pricing.
- A limited useful life creates a market risk if those products cannot be used before they become outdated.
- Between the sales channel information, the productivity data, and the supply chain data, it is possible to make more precise predictions about inventory requirements.

V.d. Distribution Analysis

- Imagine that your company has a large number of retail outlets, a smaller number of regional warehouses, and a very small number of factories.
- The optimal distribution model would arrange for the delivery of the exact number of products from each factory to its closest warehouses.
- so that each warehouse could deliver the exact number of products to each of the retail stores.
- If a company can predict demand for specific products within certain areas, though, the managers cannot only distribute the product to the right locations in the right quantities.
- Also, minimize shipping costs between factories and warehouses, and similarly between warehouses and retail stores.

Business Intelligence Applications

VI. Behavior Analysis

- This type of analytical processing makes use of historical data to look for behavior patterns that take place before the significant event.
- This allows for the following kinds of analytics:

VI.a. Purchasing Trends

- Although many product life-cycles can easily be predicted and charted, there are apparent nonlinear trends that avoid predictability
- The most notable cases are cloth sales during Dussehra and Diwali.
- Behavior analytics can be used to identify purchasing patterns that indicate a growing trend that can be used to adjust a company's productivity to customer trends.

VI.b. Web Activity

- In the world of e-commerce, the ability to draw and maintain customers to a Web site and then encourage them to commit to purchasing products.
- Different kinds of content presentation may lead different kinds of consumers to behave differently.

Business Intelligence Applications

- It is interesting to identify patterns that lead to business success (e.g., product purchase). It is called as ”success patterns”.
- Present personalization information through customer analytics should improve the probability of making a sale.

VI.c. Fraud and Abuse Detection

- Fraudulent (or abusive) behavior appears frequently in behavior patterns.
- Behavior analytics can be used to discover and eliminate patterns of suspicious behavior by provider, geographical region, agent, etc.

VI.d. Customer Attrition

- In competitive industries, it is much easier to convince a customer to stay with the company before the decision has been made to leave rather than afterwards.

- Therefore, it is important to recognize the signs that a customer is prepared to stop doing business with you.
- This can be done by evaluating patterns of behavior before previous attrition (such as a history of customer service complaints).

VI.e. Social Network Analysis

- Sometimes it is important to identify relationships between specific entities within an ecosystem and to analyze their behavior as a group.
- This kind of analytical processing is valuable for law enforcement, marketing, as well as sales optimization.

Intelligence Dashboard

- A key performance indicator (KPI) is some objective measurement of an aspect of a business that is critical to the success of that business.
- It is associated with a number of different business activities, such as customer satisfaction, productivity, supply channel performance, and profitability.
- In fact a large number of KPls can be defined in terms of measuring performance associated with many of the BI analytics functions.
- BI provide a visualization dashboard that reflects those KPI measurements.
- An intelligence dashboard will not only provide a real-time presentation of the selected KPIs but will also hook directly into the BI components that allow for further analysis.
- The following are some example KPls:

1. Regional sales figures by sales location
2. Real-time supply chain reports by supplier
3. Customer satisfaction measurements
4. Personnel statistics
5. Factory productivity
6. Average customer profitability

Business Intelligence Adds Value

- Knowledge derived from a company's data can be used as an asset.
- An investment in turning data into actionable knowledge can have a significant payoff.
- It is important to recognize that this problem cannot be solved solely by the application of technology.
- It requires senior-level management commitment to exploit discovered knowledge and have a way to measure the value of those activities.
- There are a number of BI analytics that provide business value.
- Selecting and integrating these analytics functions relies on having the right information infrastructure, as well as proper reporting and visualization.

ALM - II : Video Synthesis

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Video Synthesis

Question: Consider a global manufacturing company that faces challenges in optimizing its supply chain operations. The company wants to leverage business intelligence (BI) to improve demand forecasting, inventory management, and supplier performance. How can the company implement a BI strategy to address these challenges, and what Key Performance Indicators (KPIs) should be tracked to measure success? Analyze the potential impact of BI on the company's supply chain efficiency and decision-making processes.

Note: Watch videos to understand supply chain management and how BI optimizes the process. Understand and write answer on your ALM book.

ALM - II: Video Synthesis

Rubrics

8M - 10M	5M - 7M	1M - 4M
Excellent analysis, brilliant synthesis and accurate description.	Good analysis, but a little verbose.	Inability to either analyse or synthesise correctly or able to articulate well.

Videos

Supply Chain: What is Supply Chain Management? Definition, Introduction & Process
`https://www.youtube.com/watch?v=lZPO5RclZEo`

Supply Chain: What Is Supply Chain Management?
`https://www.youtube.com/watch?v=Lpp9bHtPAN0`

Business Intelligence: Effective BI for logistics and supply chain.
`https://www.youtube.com/watch?v=wlC-8qlY1Y0`

Books & E-Resources

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Books & E-Resources

Textbooks

1. *Loshin, D. (2012). Business intelligence: the savvy manager's guide. Newnes.*

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1. *Kemper, H. G., Mehanna, W., & Unger, C. (2010). Business Intelligence-Grundlagen und praktische Anwendungen (Vol. 3). Wiesbaden: Vieweg+ Teubner.*
2. *Moss, L. T., & Atre, S. (2003). Business intelligence roadmap: the complete project lifecycle for decision-support applications. Addison-Wesley Professional.*
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About Me

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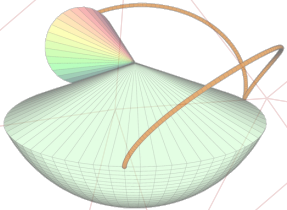

The end!